

Mission Integrity System (MIS) — V1

Scenario: Sensor Disruption

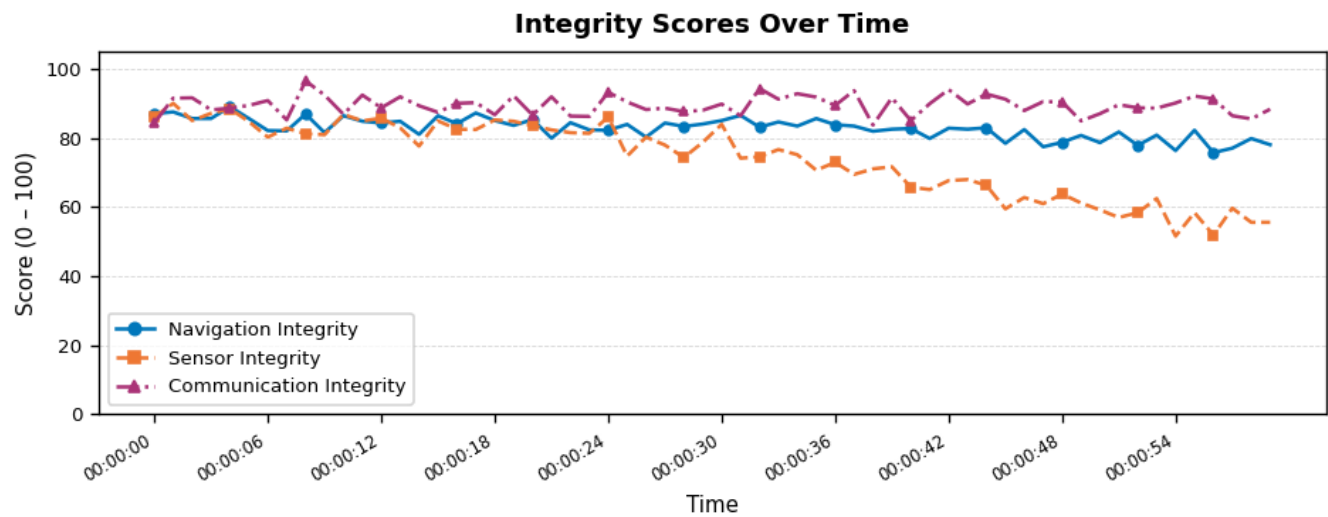
Generated: 2026-04-21 16:49:13 UTC

Demonstration report generated from simulated data

Scenario Summary

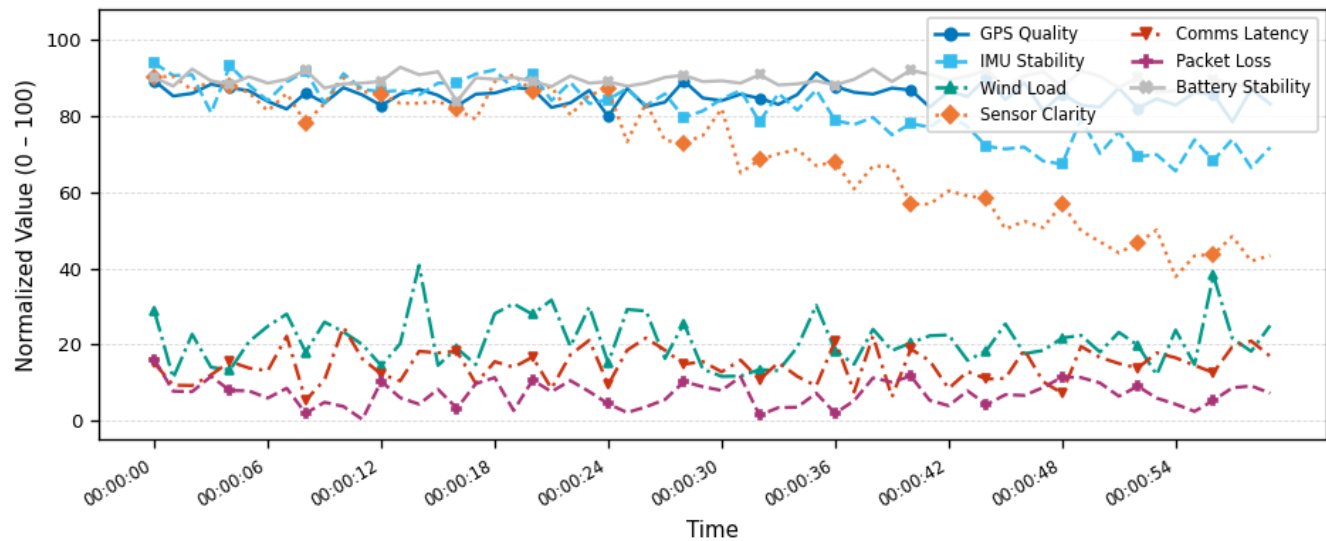
Field	Value
Final State	Degraded
Navigation Integrity	78.1 / 100
Sensor Integrity	55.6 / 100
Communication Integrity	88.4 / 100
Causal Explanation	Degraded: Sensor integrity in degraded band (40–59).

Integrity Scores Over Time



Raw Signal Channels Over Time

Raw Signal Channels Over Time



Event Log

Time	Severity	Type	Message
00:00:00	INFO	State Change	Mission state initialized: Stable.
00:00:25	WARNING	State Change	State transition: Stable → Drift Emerging. Drift Emerging: Sensor integrity in drift-emerging band (60–74).
00:00:25	WARNING	Threshold Breach	Sensor integrity crossed below threshold 75: 86.1 → 74.8.
00:00:25	WARNING	Integrity Drop	Sensor integrity dropped 11.3 points in one step: 86.1 → 74.8.
00:00:26	INFO	State Change	State transition: Drift Emerging → Stable. Stable: all integrity domains within acceptable bounds.
00:00:28	WARNING	State Change	State transition: Stable → Drift Emerging. Drift Emerging: Sensor integrity in drift-emerging band (60–74).
00:00:28	WARNING	Threshold Breach	Sensor integrity crossed below threshold 75: 78.0 → 74.4.
00:00:29	INFO	State Change	State transition: Drift Emerging → Stable. Stable: all integrity domains within acceptable bounds.
00:00:31	WARNING	State Change	State transition: Stable → Drift Emerging. Drift Emerging: Sensor integrity in drift-emerging band (60–74).
00:00:31	WARNING	Threshold Breach	Sensor integrity crossed below threshold 75: 84.0 → 74.2.
00:00:35	WARNING	Threshold Breach	Sensor integrity crossed below threshold 75: 75.2 → 70.7.
00:00:38	WARNING	Integrity Drop	Communication integrity dropped 10.0 points in one step: 93.8 → 83.8.
00:00:45	WARNING	State Change	State transition: Drift Emerging → Degraded. Degraded: Sensor integrity in degraded band (40–59).
00:00:45	WARNING	Threshold Breach	Sensor integrity crossed below threshold 60: 66.3 → 59.5.
00:00:46	WARNING	State Change	State transition: Degraded → Drift Emerging. Drift Emerging: Sensor integrity in drift-emerging band (60–74).
00:00:50	WARNING	State Change	State transition: Drift Emerging → Degraded. Degraded: Sensor integrity in degraded band (40–59).
00:00:50	WARNING	Threshold Breach	Sensor integrity crossed below threshold 60: 61.2 → 59.2.
00:00:54	WARNING	Threshold Breach	Sensor integrity crossed below threshold 60: 62.5 → 51.6.
00:00:54	WARNING	Integrity Drop	Sensor integrity dropped 10.9 points in one step: 62.5 → 51.6.

Data Excerpt (Last 15 Steps)

Showing the final 15 time steps of enriched scenario output. Scores are on a 0–100 scale.

Time	State	Nav	Sensor	Comms	GPS Qual	IMU Stab	Wind	Clarity	Latency	Pkt Loss	Batt Stab
00:00:45	Degraded	78.5	59.5	91.3	84.4	71.4	25.4	50.3	11.1	6.8	87.4
00:00:46	Drift Emergi	80.5	62.8	88.0	88.8	71.9	17.6	52.4	18.6	6.6	90.6
00:00:47	Drift Emergi	77.5	61.0	90.6	81.5	68.2	18.4	50.7	10.2	8.7	91.7
00:00:48	Drift Emergi	79.8	63.8	90.5	85.9	67.4	21.7	56.9	7.2	11.4	87.7
00:00:49	Drift Emergi	80.8	61.2	85.0	82.9	79.3	22.4	49.9	19.5	11.3	91.9
00:00:50	Degraded	78.7	59.2	87.1	82.4	70.2	18.0	47.0	16.6	9.9	90.5
00:00:51	Degraded	81.8	57.0	89.7	87.4	75.9	23.2	44.1	15.0	6.4	87.5
00:00:52	Degraded	77.9	58.4	88.8	82.1	69.4	19.8	46.6	13.8	9.0	90.1
00:00:53	Degraded	80.9	62.5	88.7	84.6	69.9	12.0	50.1	17.8	5.9	86.3
00:00:54	Degraded	76.4	51.6	90.2	82.9	65.6	23.8	37.8	16.5	4.3	86.7
00:00:55	Degraded	82.3	58.4	92.2	86.2	73.8	14.7	43.3	14.5	2.4	86.1
00:00:56	Degraded	75.8	51.9	91.4	85.8	68.4	38.2	43.7	12.6	5.3	89.7
00:00:57	Degraded	77.1	59.7	86.5	78.5	73.9	21.6	48.4	19.4	8.6	88.4
00:00:58	Degraded	79.9	55.6	85.6	87.1	66.5	18.2	41.9	20.9	9.1	87.6
00:00:59	Degraded	78.1	55.6	88.4	83.2	71.8	25.0	43.4	16.9	7.2	88.0

Limitations and Non-Goals

Mission Integrity System (MIS) V1 uses simulated telemetry-like data for demonstration and portfolio purposes only. It is not a UAV control system, flight application, autopilot, navigation engine, targeting system, safety-certified tool, predictive model, or production deployment. MIS V1 performs deterministic classification only and does not recommend or automate operational decisions.